# Baseline Results

On-demand weight re-layout | Llama3-8B FP16 linear weights

## 1. Serialized DRAM traffic

Read the PIM-placed source and write a conventional destination as separate DRAMsim3 stages. Each stage completed the same number of 128-byte requests. The table gives simulator completion upper brackets from the archived 256-cycle-resolution searches.

| Weight class | Requests / stage | Source READ (µs) | Destination WRITE (µs) | Serial sum (µs) |
|---|---|---|---|---|
| Q/O | 262,144 | 736.000 | 165.920 | 901.920 |
| K/V | 65,536 | 184.560 | 41.120 | 225.680 |
| FFN gate/up | 917,504 | 2,575.300 | 581.910 | 3,157.210 |
| FFN down | 917,504 | 4,826.080 | 581.910 | 5,407.990 |

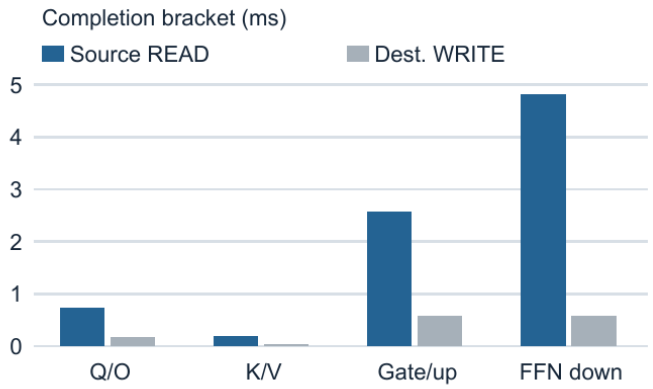



Fig. 1. Source and destination stages on a common linear scale. Exact values are given above.

Across 32 blocks, applying multiplicities 2, 2, 2, 1 gives 447.284 ms for all 224 modeled linear matrices. FFN down and gate/up both occupy 112 MiB. Their destination times are equal, while FFN-down source reads take 1.874× as long. The replay does not isolate the contribution of placement from access order.

## 2. Analytical prefill reference

| Prompt tokens | Ideal SoC projection (ms) | Serialized copy (ms) | Baseline proxy (ms) |
|---|---|---|---|
| 8 | 68.362 | 447.284 | 515.646 |
| 16 | 68.567 | 447.284 | 515.851 |
| 32 | 68.977 | 447.284 | 516.260 |
| 64 | 69.796 | 447.284 | 517.079 |
| 128 | 71.434 | 447.284 | 518.718 |

The proxy adds the ideal SoC projection estimate to the serialized traffic component before rounding. Host rearrangement, overlap, attention, and runtime are outside this calculation. Sources: verified_baseline_*_2k/summary.json and facil_step6_comparison/ttft_comparison.csv.